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MILITARY SPECIFICATION

FILTER, LOWPASS (NAVORD Dwg 1563732)
(FOR TORPEDO MARK 37 MODS 0, 2, and 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements for the procurement of a Lowpass Filter, NAVORD Drawing 1563732, which is used in the enabler assembly of an acoustic panel.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-621	Box, Wood, Nailed and Lock-corner

Military

MIL-R-11	Resistors, Fixed, Composition (Insulated), General Specification for
MIL-P-116	Preservation, Methods of
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

FSC 1355

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

Naval Ordnance Systems Command

NAVORD DWG 1563732	Filter, Lowpass

(Copies of specifications, standards, drawings, and publications requested by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

UNITED STATES OF AMERICA STANDARDS INSTITUTE

USAS Y32.2 1967	Graphic Symbols for Electrical and Electronics Diagrams

(Application for copies should be addressed to DOD Single Stock Point, U. S. Naval Supply Depot, Philadelphia, Pennsylvania 19120.)

3. REQUIREMENTS

3.1 Description. The lowpass filter is a plug-in electronic assembly composed of resistors and capacitors which are contained in a sealed metal

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container. Its purpose is to pass a narrow band of low frequencies to the next stage while rejecting or shunting to ground all other frequencies applied to its input terminals.

3.2 General requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the lowpass filter shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing lowpass filters that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Materials. Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of the lowpass filter shall conform strictly to the detailed requirements indicated on the applicable drawings, specifications, and publications listed in NAVORD Dwg 1563732.

3.2.4 Surface treatment, construction, and dimensions. Surface treatment, construction, and dimensions for the lowpass filter shall be in accordance with the applicable drawings, specifications, and publications listed in NAVORD Dwg 1563732.

3.2.5 Moisture, fungus, and corrosion. The lowpass filter shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on the drawings and specifications listed in NAVORD Dwg 1563732.

3.2.6 Marking. Marking of assemblies, subassemblies and parts shall be in accordance with applicable drawings listed in NAVORD Dwg 1563732.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

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3.3 Symbols. Interpret schematic diagram symbols in accordance with USAS Y32.2.

3.4 Performance requirements and product characteristics. The lowpass filter shall meet the following performance requirements and product characteristics:

3.4.1 All pin designations referred to in this specification are shown on NAVORD Dwg 1563732.

3.4.2 Open circuit attenuation characteristics. The lowpass filter shall have the following attenuation characteristics when tested as specified in 4.4.4.2 of this specification. In all cases, the attenuation shall be referred to the response at 20 Hertz (Hz).

3.4.2.1 The attenuation of 55 Hz shall be 6 plus or minus 0.5 db.

3.4.2.2 The attenuation of 105 Hz shall be 12 plus or minus 0.5 db.

3.4.2.3 The attenuation of 400 Hz shall be at least 30 db.

3.4.3 Output balance. The lowpass filter output shall be 0.6 volt direct current (vdc) maximum, when measured as specified in 4.4.4.3 of this specification.

3.4.4 Direct current resistance. The direct current resistance between pins 4 and 5, and pins 4 and 6 shall be 750,000 ohms plus or minus 0.1 percent, when measured as specified in 4.4.4.4.

3.4.5 Dielectric strength. The lowpass filter shall withstand the specified potentials without breakdown, puncturing of insulation, or arcing, when tested as specified in 4.4.4.5.

3.4.5.1 The dielectric test voltage between any two mutually insulated terminals shall be 300 vdc for a period of one minute.

3.4.5.2 The dielectric test voltage between any terminal and case shall be 200 vdc for a period of one minute.

3.4.6 Insulation resistance. The lowpass filter shall meet the minimum insulation resistance requirements when tested as specified in 4.4.4.6.

3.4.6.1 The insulation resistance between any two mutually insulated terminals shall be 10,000 megohms minimum.

3.4.6.2 The insulation resistance between any terminal and the filter case shall be 10,000 megohms minimum.

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3.4.7 Operating life. The operating life of the lowpass filter, when tested as specified in 4.5.5, shall be at least 800 cycles of operation.

3.5 Environmental requirements. The lowpass filter shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.5.

3.5.1 Vibration. The lowpass filter shall meet the following vibration requirements:

3.5.1.1 Vibration, nonoperating. The lowpass filter shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.2.

3.5.1.2 Vibration, operating. The lowpass filter shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.5.1.3.

3.5.1.3 Vibration, transportation. The lowpass filter shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.4.

3.5.2 Shock, transportation. The lowpass filter shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.

3.5.3 Temperature and humidity

3.5.3.1 High temperature and humidity, nonoperating. The lowpass filter shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.1.

3.5.3.2 High temperature, operating. The lowpass filter shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.3.2.

3.5.3.3 Low temperature, nonoperating. The lowpass filter shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.3.

3.5.3.4 Low temperature, operating. The lowpass filter shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.3.4.

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3.5.4 Immersion. The lowpass filter shall show no evidence of leakage during the immersion test specified in 4.5.4 and thereafter shall meet the performance and product requirements of this specification.

3.6 Cleanup. Prior to and after final assembly all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as excessive rosin flash shall be removed.

3.7 Workmanship. The lowpass filter, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges.

3.7.1 The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the lowpass filter shall be classified as follows:

(a) Preproduction inspection (paragraph 4.5)

(b) Quality conformance inspection (paragraph 4.4)

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(c) Periodic production inspection (paragraph 4.5)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five lowpass filters, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.5, at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the lowpass filter by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.4, a random sample of five lowpass filters will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The lowpass filters shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Quality conformance. Each lowpass filter delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirements specified herein shall be considered cause for rejection.

4.4.1 Test conditions. Unless otherwise specified, the lowpass filter shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18.3 degrees centigrade (C) to 35 degrees C).

4.4.1.2 Altitude: Normal ground.

4.4.1.3 Vibration: None.

4.4.1.4 Humidity: Room ambient to 95 percent relative maximum.

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4.4.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests.

4.4.3 Test equipment

4.4.3.1 The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.4.3.1.1 Audio oscillator. TS-421A/U or equivalent.

4.4.3.1.2 Electronic counter. Hewlett-Packard 522B or equivalent.

4.4.3.1.3 Multimeter. ME-6B/U or equivalent.

4.4.3.1.4 Resistors. 10 megohms plus or minus 5 percent, 1/2 watt; MIL-R-11 designation RC20GF106J, two required.

4.4.3.1.5 Voltmeter, dc. Senior Voltohm-yst, RCA WV-97A or equivalent.

4.4.3.1.6 Battery, dry. Capable of furnishing 300 vdc plus or minus 1 percent for load variations up to one milliampere.

4.4.3.1.7 Insulation tester. Radio frequency laboratories 184C or equivalent.

4.4.3.1.8 Resistance bridge capable of measuring 750,000 ohms to 0.1 percent.

4.4.4 Test procedure. The following tests constitute the quality conformance tests of the lowpass filter.

4.4.4.1 Unless otherwise specified, all quality conformance tests shall be conducted with the test equipment of 4.4.3 and the test circuits contained in this specification. The test circuits shall be connected as described in the following paragraphs:

4.4.4.2 To measure open-circuit attenuation characteristics as set forth in 3.4.2:

4.4.4.2.1 Preparation for measurements. Connect the lowpass filter as shown in Figure 1, ensuring that the ground terminals of the following test equipment are connected to Point G:

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(a) Audio oscillator (4.4.3.1.1)

(b) Electronic counter (4.4.3.1.2)

(c) Multimeter (4.4.3.1.3)

Careful placement of the leads is necessary to prevent pickup.

4.4.4.2.2 Set the oscillator IMPEDANCE switch to the 5000-ohm position.

4.4.4.2.3 Set the oscillator LOAD switch to ON.

4.4.4.2.4 Check the calibration of the counter in accordance with the instructions provided.

4.4.4.2.5 When the counter has been satisfactorily checked, set the counter controls to measure frequency.

4.4.4.2.6 Connect the oscillator and counter to Point A of Figure 1.

4.4.4.2.7 Connect the multimeter probe to Point C of Figure 1.

4.4.4.2.8 Adjust the oscillator frequency until the counter indicates a frequency of 20 plus or minus 1 Hz; then adjust the oscillator AMPLITUDE control for a voltage that is at least 30 db above the minimum voltage than can be indicated on the multimeter.

4.4.4.2.9 Record the voltage indicated on the multimeter.

4.4.4.2.10 Remove the multimeter probe from Point C and connect it to Point A.

4.4.4.2.11 Record the voltage indicated on the multimeter.

4.4.4.2.12 Adjust the oscillator frequency until the counter indicates a frequency of 55 plus or minus 1 Hz, then adjust the oscillator AMPLITUDE control until the multimeter indicates within 0.1 db the voltage recorded in 4.4.4.2.11.

4.4.4.2.13 Remove the multimeter probe from Point A and connect it to Point C. The voltage indicated on the multimeter shall be 6 plus or minus 0.5 db below the voltage recorded in 4.4.4.2.9.

4.4.4.2.14 Remove the multimeter probe from Point C and connect it to Point A.

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4.4.4.2.15 Adjust the oscillator frequency until the counter indicates a frequency of 105 plus or minus 2 Hz; then adjust the oscillator AMPLITUDE control until the multimeter indicates within 0.1 db the voltage recorded in 4.4.4.2.11.

4.4.4.2.16 Remove the multimeter probe from Point A and connect it to Point C.

4.4.4.2.17 The voltage indicated on the multimeter shall be 12 plus or minus 0.5 db below the voltage recorded in 4.4.4.2.9.

4.4.4.2.18 Remove the multimeter probe from Point C and connect it to Point A.

4.4.4.2.19 Adjust the oscillator frequency until the counter indicates a frequency of 400 plus or minus 3 Hz; then adjust the oscillator AMPLITUDE control until the multimeter indicates within 0.1 db the voltage recorded in 4.4.4.2.11.

4.4.4.2.20 Remove the multimeter probe from Point A and connect it to Point C.

4.4.4.2.21 The voltage indicated on the multimeter shall be at least 30 db below the voltage recorded in 4.4.4.2.9.

4.4.4.2.22 Connect the oscillator and counter probes to Point B of Figure 1.

4.4.4.2.23 Connect the multimeter probe to Point D of Figure 1.

4.4.4.2.24 Adjust the oscillator frequency until the counter indicates a frequency of 20 plus or minus 1 Hz; then adjust the oscillator AMPLITUDE control until the multimeter indicates a voltage at least 30 db above the minimum voltage that can be indicated on the multimeter.

4.4.4.2.25 Record the voltage indicated on the multimeter.

4.4.4.2.26 Remove the multimeter probe from Point D and connect it to Point B.

4.4.4.2.27 Record the voltage indicated on the multimeter.

4.4.4.2.28 Adjust the oscillator frequency until the counter indicates a frequency of 55 plus or minus 1 Hz; then adjust the oscillator AMPLITUDE control until the multimeter indicates within 0.1 db the volage recorded in 4.4.4.2.27.

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4.4.4.2.29 Remove the multimeter probe from Point B and connect it to Point D. The voltage indicated on the multimeter shall be 6 plus or minus 0.5 db below the voltage recorded in 4.4.4.2.25.

4.4.4.2.30 Remove the multimeter probe from Point D and connect it to Point B.

4.4.4.2.31 Adjust the oscillator frequency until the counter indicates a frequency of 105 plus or minus 2 Hz; then adjust the oscillator AMPLITUDE control until the multimeter indicates within 0.1 db the voltage recorded in 4.4.4.2.27.

4.4.4.2.32 Remove the multimeter probe from Point B and connect it to Point D. The voltage indicated on the multimeter shall be 12 plus or minus 0.5 db below the voltage recorded in 4.4.4.2.25.

4.4.4.2.33 Remove the multimeter probe from Point D and connect it to Point B.

4.4.4.2.34 Adjust the oscillator frequency until the counter indicates a frequency of 400 plus or minus 3 Hz; then adjust the oscillator AMPLITUDE control until the multimeter indicates within 0.1 db the voltage recorded in 4.4.4.2.27.

4.4.4.2.35 Remove the multimeter probe from Point B and connect it to Point D. The voltage indicated on the multimeter shall be at least 30 db below the voltage recorded in 4.4.4.2.25.

4.4.4.2.36 This completes the open-circuit attenuation measurements.

4.4.4.2.37 When the attenuation measurements have been satisfactorily completed, proceed to the output balance measurements.

4.4.4.3 To measure the output balance as set forth in 3.4.3:

4.4.4.3.1 Connect the lowpass filter to the battery (4.4.3.1.6) as shown in Figure 2.

4.4.4.3.2 Connect the voltmeter (4.4.3.1.5) to Points A and B of Figure 2; the positive terminal connected to Point A and the negative terminal to Point B.

4.4.4.3.3 Adjust the battery voltage until the voltmeter indicates a voltage of 300 plus or minus 3.0 volts direct current (vdc).

4.4.4.3.4 Remove the voltmeter terminals From points A and B.

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4.4.4.3.5 Connect the voltmeter terminals to pins 1 and 2 of the lowpass filter.

4.4.4.3.6 The voltage indicated on the voltmeter shall not exceed 0.6 vdc.

4.4.4.4 To measure the dc resistance as set forth in 3.4.4:

4.4.4.4.1 Connect one terminal of the resistance bridge (4.4.3.1.8) to pin 4 of the lowpass filter and the other terminal to pin 5. The resistance shall be 750,000 ohms plus or minus 0.1 percent.

4.4.4.4.2 Connect one terminal of the resistance bridge (4.4.3.1.8) to pin 4 of the lowpass filter and the other terminal to pin 6. The resistance shall be 750,000 ohms plus or minus 0.1 percent.

4.4.4.5 To check dielectric strength as set forth in 3.4.5:

4.4.4.5.1 Connect the insulation tester (4.4.3.1.7) terminals to pins 1 and 9 of the lowpass filter and apply a potential of 300 plus or minus 10 vdc for one minute. There shall be no breakdown or arcing during this one minute period (the neon lamp on the insulation tester shall not light). Upon completion of this check, remove the terminals from the lowpass filter.

4.4.4.5.2 Connect the insulation tester terminals to pins 2 and 3 of the lowpass filter and apply a potential of 300 plus or minus 10 vdc for one minute. There shall be no breakdown or arcing during this one-minute period. Upon completion of this check, remove the terminal connections from the lowpass filter.

4.4.4.5.3 Connect the insulation tester terminals to the lowpass filter case (mounting stud) and all terminals (pins 1 through 9 connected in parallel) and apply a potential of 200 plus or minus 10 vdc for one minute. There shall be no breakdown or arcing during this one-minute period. Upon completion of this check remove the terminal connections from the lowpass filter.

4.4.4.6 To measure the insulation resistance as set forth in 3.4.6:

4.4.4.6.1 Connect the insulation tester terminals to pins 1 and 9 and apply a potential of 100 vdc for two minutes. Immediately after this two minute period, the resistance at the test potential shall be 10,000 megohms minimum.

4.4.4.6.2 Connect the insulation tester terminals to pins 2 and 3 and apply a potential of 100 vdc for two minutes. Immediately after this

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two-minute period, the resistance at the test potential shall be 10,000 megohms minimum.

4.4.4.6.3 Apply 100 volts to each of pins 1 through 9 in turn with respect to the case (mounting stud) for two minutes each. At the end of each two-minute period the resistance reading between the individual pin being tested and the case shall be 10,000 megohms minimum.

4.4.4.7 Visual and dimensional examination. Each lowpass filter shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.4.4.8 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

4.5 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.5.1 Vibration

4.5.1.1 Mounting. The lowpass filter, less pack or packaging, shall be mounted to a test fixture as shown in Figure 3 and tested as specified in 4.5.1.2 and 4.5.1.3.

4.5.1.2 Vibration, nonoperating. The lowpass filter shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.1.1.

4.5.1.3 Vibration, operating. The lowpass filter shall be subjected to the "Vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.5.1.2.

4.5.1.4 Vibration, transportation. The lowpass filter shall be subjected to the "Vibration, transportation" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.1.3.

4.5.2 Shock, transportation. The lowpass filter shall be subjected to the "Shock, transportation" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.2.

4.5.3 Temperature and humidity

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4.5.3.1 High temperature and humidity, nonoperating. The lowpass filter, shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.3.1.

4.5.3.2 High temperature, operating. The lowpass filter shall be subjected to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.5.3.2.

4.5.3.3 Low temperature, nonoperating. The lowpass filter shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.3.3.

4.5.3.4 Low temperature, operating. The lowpass filter shall be subjected to the "Low temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.5.3.4.

4.5.4 Immersion. The lowpass filter shall show no evidence of leakage while being subjected to the immersion test specified in MIL-STD-202, Method 104, Condition A, and thereafter shall meet the performance and product requirements of this specification.

4.5.5 Operating life. The test for operating life shall be conducted as follows:

4.5.5.1 Connect the lowpass filter as shown in Figure 2. Apply the 300 vdc for 30 plus or minus 1 minute, then allow the circuit to rest for at least one hour. An electronic power supply may be used in place of the battery for this test only, if desired. This completes one cycle of operation.

3.5.5.2 Repeat 4.5.5.1 until 800 cycles of operation are completed, after which the lowpass filter shall meet all the requirements of 3.4.1 through 3.4.6.

4.6 Waiver of tests. Tests specified in 4.5.1.4 and 4.5.2 may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.7 Functional tests. At the conclusion of the tests specified in 4.5.1.4 and 4.5.2, functional and other tests, as applicable, shall be conducted on the lowpass filter to determine freedom from operational malfunction and compliance with all performance requirements and product characteristics of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

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5.1.1 Level A

5.1.1.1 Unit packaging. The filter shall be packaged in accordance with Method IIB of MIL-P-116.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The filter shall be packaged in accordance with Method III of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Consolidation container. The filter packaged as described in 5.1.1.1 shall be packed in a wooden box in accordance with PPP-B-621, Class 2.

5.2.1.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 101 (MS 53102) (see 6.2).

5.2.2 Level C. The filter, packaged in accordance with 5.1.3.1, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Levels A or C cushioning. The unit package shall incorporate sufficient cushioning material, bracing or other adequate shock absorbing devices to ensure that the filter will meet all the requirements of Section 3 after being tested in accordance with 4.5.1.4 (Vibration) and 4.5.2 (Shock). At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent.

5.4 Marking

5.4.1 Special markings. None, unless otherwise specified.

5.4.2 Normal markings. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

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6. NOTES

6.1 Intended use. The lowpass filter is intended for use in Torpedo Mark 37 Mods 0, 2, and 3.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) Data required other than that specified in 3.2.1.

(g) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

(h) Guarantee clause (see 6.4).

6.3 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

6.3.1 Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

6.3.2 Level C. This level shall be used only when it is definitely known that the packaged items are to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Guarantee. The contract or order shall include a requirement for a guarantee stating that any lowpass filter which fails to meet the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

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6.5 The contract or order shall include a sufficient number of extra lowpass filters to cover preproduction and periodic production inspection samples.

Preparing Activity
Navy - OS
Project No. 1355 - N213

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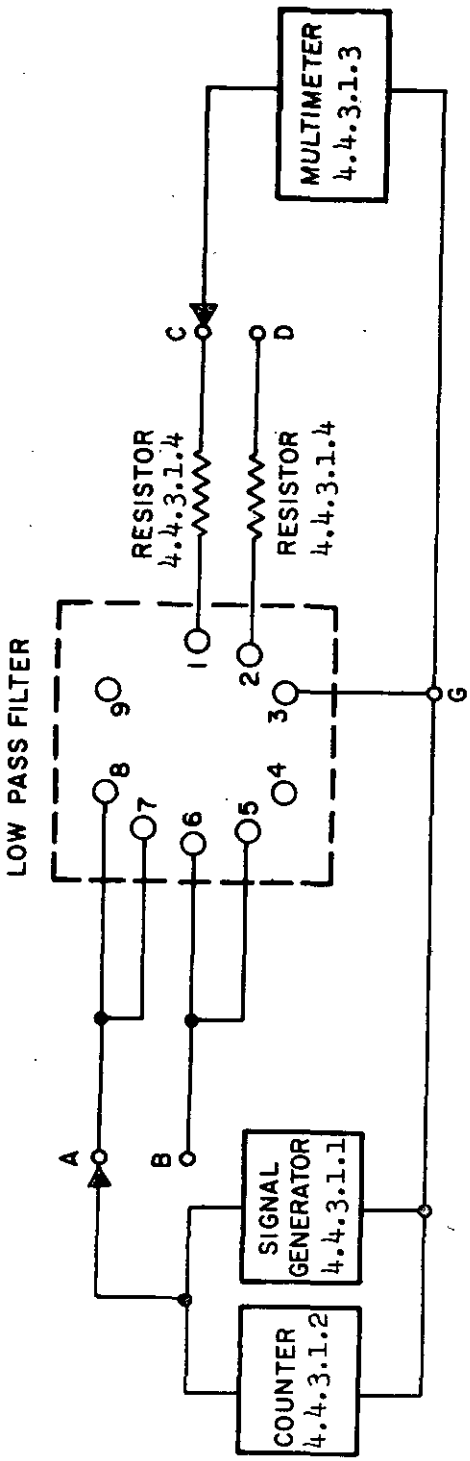


Figure 1

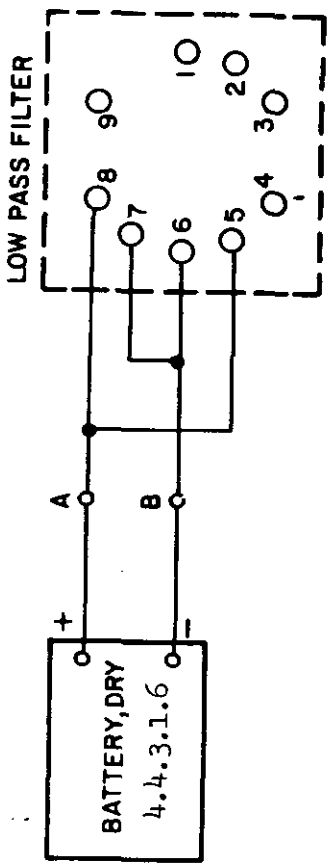
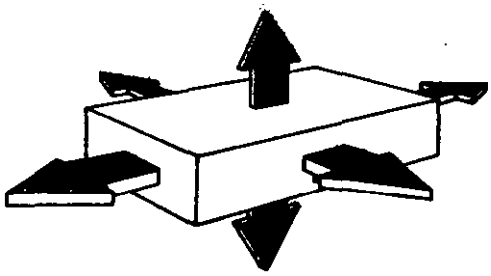


Figure 2

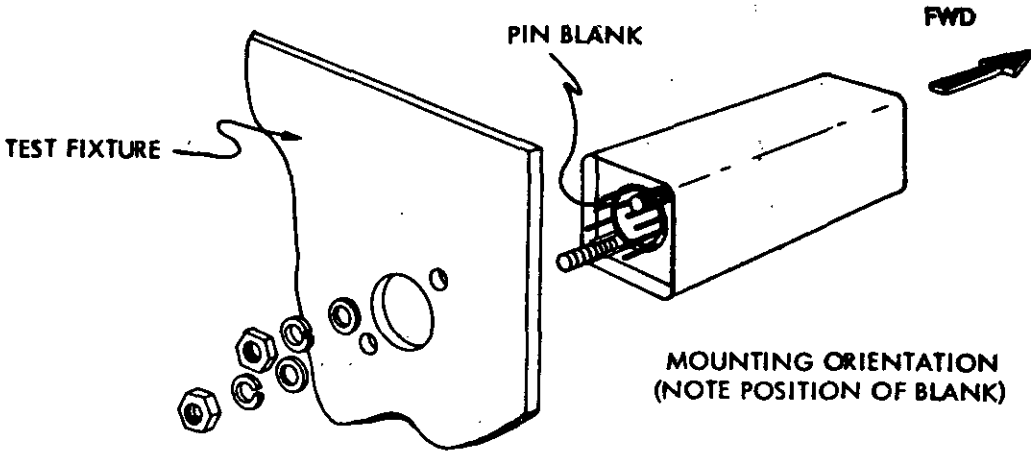
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FIGURE 3. FILTER NAVORD DWG 1563732
MOUNTING REQUIREMENTS



DIRECTIONS OF VIBRATION INPUTS





NOTES:

1. FOR VIBRATION, ORIENTATION OF TEST ITEM WITH RESPECT TO EARTH'S GRAVITATIONAL FIELD SHALL BE MAINTAINED FOR ALL TEST DIRECTIONS.
2. FOR VIBRATION, THE TEST ITEM SHALL BE SECURED TO THE TEST FIXTURE AS SHOWN.
3. THE TEST FIXTURE SHALL NOT ALTER THE INPUTS FROM THE VIBRATION EXCITER TO THE MOUNTING POINTS OF THE TEST ITEM.

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